

RESEARCH INTEREST

- Machine Learning, Optimization Theory, Operations Research

EDUCATION

- **Stanford University** **California, US**
Management Science and Engineering *Sep. 2026–Jul. 2031(Expected)*
- **Peking University** **Beijing, China**
B.A. in Mathematics, School of Mathematical Sciences *Jun. 2023–Jul. 2026*
- **Peking University** **Beijing, China**
B.A. in Chemistry, College of Environmental Sciences and Engineering *Sep. 2022–Jun. 2023*

SELECTED HONORS AND AWARDS

- **Finalist in INFORMS Applied Probability Society Best Student Paper Prize, 2025**
Annual award recognizing outstanding student research in applied probability [Link] [Award] *Oct. 2025*
- **Applied Mathematics Elite Program**
The program lead by Prof. [Weinan E](#) accepted only 15 people this year *Jun. 2024*
- **Silver Medal, 35th Chinese Chemical Olympiad (National Final)**
The prize is awarded to the top 150 high school students in chemistry throughout China *Nov. 2021*

WORKING EXPERIENCE

- **Decision Intelligence Lab, DAMO Academy** **Hangzhou, China**
Research Intern *Jul. 2025–Sep. 2025*
 - Developed the complete codebase for MetaFlow and built execution sandboxes (including tool calling like RAG search and Lean4 theorem proving) for online generated workflow and verification, enabling real-time execution feedback for reinforcement learning with verifiable rewards.

RESEARCH EXPERIENCE ON LARGE LANGUAGE MODELS AND OPERATIONS RESEARCH

- **Online Scheduling on LLM Inference** **Massachusetts Institute of Technology**
Advisor: Prof. David Simchi-Levi, Massachusetts Institute of Technology *Oct. 2024 – May. 2025*
 - **Optimizing LLM Inference: Fluid-Guided Online Scheduling with Memory Constraints**
 $(\alpha\text{-}\beta)$ [Ruicheng Ao*](#), [Gan Luo*](#), [David Simchi-Levi](#), [Xinshang Wang](#) [SSRN] [Arxiv] [Code1] [Code2] [Demo]
 - * **Major Revision at Operations Research.**
 - * Preliminary version accepted to **NeurIPS 2025 MLxOR Workshop.**
 - * **Finalist in INFORMS Applied Probability Society Best Student Paper Prize, 2025.** [Link] [Award]
 - * Formulated the LLM inference as a multi-stage online scheduling task with stochastic queueable requests, proposed a novel online batching algorithm for LLM inference and proved that the algorithm achieves near-optimal throughput while controlling latency and Time to First Token (TTFT).
 - * Conducted numerical experiments on synthetic and real-world datasets with Llama-7B on A100 GPU to validate theoretical results.
- **Analysis of batching and scheduling algorithms in LLM inference** **Columbia University**
Advisor: Prof. Jing Dong, Columbia University *Apr. 2025 – Present*
 - **Work in progress, Analysis of Continuous Batching Algorithm in LLM Inference**
 - * In this work, we first formulated the continuous batching algorithm as a discrete-time model. Then we analyzed its steady state and its dynamics behavior under overloaded conditions. Next we will further analyze its dynamics under admission control.
- **LLM Agent and Workflow Generation** **DAMO Academy**
Advisor: Prof. Wotao Yin, DAMO Academy & Prof. Bin Dong, Peking University *Apr. 2025 – Present*
 - **MetaFlow: A Meta Approach of Training LLMs into Generalizable Workflow Generators**
 $(\alpha\text{-}\beta)$ [Gan Luo*](#), [Zihan Qin*](#), [Bin Dong](#), [Wotao Yin](#)

- * **Submitted, Under Review.**
- * Formulated workflow generation as a meta-learning problem where LLMs learn to compose task-level solution strategies from operators, producing reusable workflows that generalize across problem instances rather than instance-specific solutions.
- * Developed a two-stage training approach combining supervised fine-tuning on synthetic workflow data with reinforcement learning with verifiable rewards (RLVR), using execution feedback across instances to improve end-to-end success rates.
- * Demonstrated strong zero-shot generalization to untrained tasks and novel operator sets, achieving performance comparable to state-of-the-art baselines on in-domain tasks across benchmarks in question answering, code generation, and mathematical reasoning.

RESEARCH EXPERIENCE ON OPTIMIZATION THEORY

Convergence and Speedup Analysis of Distributed Optimization Algorithms

Peking University

Advisor: Prof. Kun Yuan, Peking University

Nov. 2023 – Jun. 2025

o Push-Pull Algorithm Provably Achieves Linear Speedup Over Arbitrary Network Topologies

Liyuan Liang*, **Gan Luo***, Kun Yuan [Arxiv] [Code: Linear Speedup] [Notes]

- * Submitted to **SIAM Journal on Optimization**.
- * Proposed a novel multi-step descent analysis framework and first proved that the [Push-Pull algorithm](#) achieves linear speedup over arbitrary strongly connected digraphs. Our multi-step analysis resolved the non-vanishing noise issue inherent in [traditional single-step approaches](#). Also see the [notes](#).
- * Conducted all numerical experiments to validate the linear speedup property we proved.
- o **Achieving Linear Speedup and Optimal Complexity for Decentralized Optimization over Row-stochastic Networks**

Liyuan Liang*, Xinyi Chen*, **Gan Luo***, Kun Yuan [Arxiv] [Code]

- * **Accepted to ICML 2025, Spotlight**
- * Introduced novel metrics to characterize the influence of row-stochastic mixing matrices and established the first convergence lower bound for decentralized optimization over row-stochastic networks.
- * Developed a new analysis framework proving that [PULL-DIAG](#) achieves linear speedup and proposed a multi-gossip protocol that resolves instability issues and attains the lower bound with near-optimal complexity.
- * Conducted all numerical experiments to validate the theoretical results on convergence lower bound, linear speedup, and near-optimal complexity.

SKILLS & OTHERS

- . **Computer Skills:** Python (&PyTorch), Cuda, Lean4, MATLAB, C++, LaTeX
- . **Language:** [Sichuanese dialects](#) (Native), Mandarin Chinese (Native), English (Fluent)
- . **The Test of English as a Foreign Language (TOEFL-IBT):** Total 101 with 27(R)+28(L)+22(S)+24(W)
- . **Conference Reviewing:** ICLR 2026